
Toscanini 03 buyer decision card

invbench-p03-phe-biofouling156ae5155c73 · run r401-bench-p03-phe-biofouling · maturity
EARLY_TECHNICAL_EVALUATION · 3D class CONCEPTUAL_3D

What is it?

A recorded invention candidate (inv:bench-p03-phe-biofouling:156ae5155c73) for the problem: •

Problem Id: bench-p03-phe-biofouling

- Device: stainless steel plate heat exchanger in a district cooling loop
- Failure: biofilm and scale accumulation on plate channels reduces overall heat transfer coefficient and raises pressure drop between cleanings
- Failure Mode: fouling
- Constraint: sust... — the proposed intervention: • Mechanism: Real-time distributed fiber optic sensing combined with microcapsule-based self-healing agents that release when corrosion is detected
- Intervention: Embedding microstructured optical fiber sensors and corrosion-activated microcapsules containing healing agents throughout the stainless...

What is actually established?

The architecture survived the recorded adversarial challenge, adjudication and evolution stages; status EVOLVED_INVENTION_CANDIDATE. Evidence by class: direct support: 10; partial support: 5; background: 0; analogy: 0; contradictory: 0; irrelevant: 1. Model-derived engineering content is labeled MODEL_DERIVED; nothing claims physical validation (no PHYSICAL_OBSERVATION exists in this package).

What is not established?

FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005 — 7 recorded unknowns, each with a resolution action in UNKNOWN_ROADMAP.json. No numeric design decision is possible until the blockers are sourced.

What could kill it?

the falsification rule recorded in the decisive experiment contract

Why does it matter?

reliable operation of stainless steel plate heat exchanger in a district cooling loop

What is different?

- Collision Novelty Risk: UNRESOLVED_INSUFFICIENT_EVIDENCE
- Prior Art Status: UNRESOLVED_INSUFFICIENT_EVIDENCE
- Multi Source Directions: • Discovery: (sources_recorded, sources_errored)
- Destruction: (sources_recorded, sources_errored)
- Transfer: (sources_recorded, sources_errored)
- Reality: (s...

What would we build?

- WP-01: test article first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed); measurement physical bench/clinical experiment — NOT simulation
- WP-02: test article constraint-compliance envelope article; measurement constraint-compliance

characterization

- WP-03: test article article for VF-001; measurement Accelerated aging test with simultaneous chloride exposure and mechanical stress while monitoring healing agent depletio...
- WP-04: test article article for VF-002; measurement run killer experiment: falsification_test_from_candidate (EIG/cost 0.4557178723356231)
- WP-05: test article article for VF-003; measurement temperature-rise mapping at max duty
- WP-06: test article article for VF-004; measurement temperature-rise mapping at maximum duty with IR or embedded sensing (measured quantity: temperature rise)
- WP-07: test article article for VF-005; measurement thermal-resistance trend over accelerated cycling (measured quantity: mechanical stress / fatigue endurance)
- WP-08: test article article for VF-006; measurement temperature-rise mapping at max duty
- WP-09: test article article for VF-007; measurement worst-case ambient soak with staged power steps and stability monitoring (measured quantity: power / energy deposition)

What must the buyer bring?

canonical problem context: INDUSTRIAL_PROCESS

buyer type: industrial process-equipment / component manufacturer (candidate buyer class for an industrial process technology) [APPLICABLE]

engineering capability: process/mechanical engineering with the domain's physics (see the engineering specification's governing model) [APPLICABLE]

engineering capability: field/plant integration engineering for the recorded operating environment [APPLICABLE]

manufacturing capability: general industrial fabrication (machining, welding/brazing, sheet/plate work, assembly) — the specific processes are the domain candidates recorded in the engin... [APPLICABLE]

manufacturing capability: quality control appropriate to the recorded failure modes [APPLICABLE]

regulatory: applicable industrial codes/standards (e.g. pressure equipment, piping, electrical)

UNKNOWN until determined for the installation jurisdiction — the buyer's regulatory affairs makes the determination; medical-device regulation is NOT_APPLIC... [UNKNOWN]

medical regulatory: medical-device regulatory requirements (FDA 510(k)/PMA, ISO 10993

biocompatibility, ISO 13485 medical QMS) [NOT_APPLICABLE]

market channels: industrial process / water-utility / plant engineering channels appropriate to the recorded application [APPLICABLE]

What is the cheapest decisive next experiment?

physical bench/clinical experiment — NOT simulation

What should we do next?

Fund and run the recorded decisive experiment (physical bench/clinical experiment — NOT simulation) — the pre-registered rule either kills the architecture or graduates it to a sourced measurement.

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.